

Hyperparameters Tuning (CSP + CNN Baseline)

#	Component	Parameter Name	Description	Current	Sweep values
1	Preprocessing	Filter Bandpass	Frequency bands (Hz) to extract motor imagery (alpha/mu & beta).	8 - 30 Hz	8-30, 8-35, 7-30, 4-38
2	Preprocessing	Epoch Window	Start/end time (s) relative to cue. Crops noise from transitions.	2.0 - 6.0 s	2.0-6.0, 2.5-5.5, 2.0-5.0
3	Preprocessing	Channel Count	Number of spatial channels selected from the 22 GDF channels.	10	8, 10, 12, 16, 22
4	CSP Space	CSP_Components	Number of spatial filters extracted by CSP to project signals.	4	4, 6, 8
5	CSP Space	Regularization	Shrinkage estimation used when trial count is scarce.	None	None, Ledoit-Wolf, OAS
6	CNN Classifier	Temp. Kernel	Width of initial temporal filter (samples). Affects frequency coverage.	64	32, 64, 128
7	CNN Classifier	Depth_Multiplier	Spatial filters learned per temporal filter in DepthwiseConv2D.	2	1, 2, 4
8	CNN Classifier	Dropout Rate	Regularization to prevent overfitting on small datasets.	0.5	0.3, 0.4, 0.5, 0.6
9	CNN Classifier	Learning_Rate	Step size for the Adam optimizer during backpropagation.	0.001	1e-4, 5e-4, 1e-3, 2e-3
10	CNN Classifier	Batch_Size	Number of samples processed per step. Smaller adds noise effect.	16	8, 16, 32, 64

Sweep 1 :

baseline_parameter_sweep.py

csp_regs = [None, 'ledoit_wolf']

time_windows = [(2.0, 6.0), (2.5, 5.5)]

kernel_lengths = [64, 128]

dropout_rates = [0.4, 0.5]

Results :

Config_Index	CSP_Reg	Time_Window	Kernel_Length	Dropout_Rate	Avg_Accuracy (baseline : real only)
1 curr	None	(2.0, 6.0)	64	0.4	0.6933
2	None	(2.0, 6.0)	64	0.5	0.6895
3	None	(2.0, 6.0)	128	0.4	0.6887
4	None	(2.0, 6.0)	128	0.5	0.6848
5	None	(2.5, 5.5)	64	0.4	0.6373
6	None	(2.5, 5.5)	64	0.5	0.6559
7	None	(2.5, 5.5)	128	0.4	0.6319
8	None	(2.5, 5.5)	128	0.5	0.6273
9	ledoit_wolf	(2.0, 6.0)	64	0.4	0.6872
10	ledoit_wolf	(2.0, 6.0)	64	0.5	0.6818
11	ledoit_wolf	(2.0, 6.0)	128	0.4	0.6741
12	ledoit_wolf	(2.0, 6.0)	128	0.5	0.6686
13	ledoit_wolf	(2.5, 5.5)	64	0.4	0.6227
14	ledoit_wolf	(2.5, 5.5)	64	0.5	0.6451
15	ledoit_wolf	(2.5, 5.5)	128	0.4	0.6319
16	ledoit_wolf	(2.5, 5.5)	128	0.5	0.6312

Sweep 2 :

Baseline_parameter_sweep2.py

Best of 2 runs of CNN

```
csp_components = [4, 6]      # n_components for CSP  
depth_multipliers = [1, 2]  # DepthwiseConv2D depth multiplier  
learning_rates = [1e-3, 5e-4] # learning rate for Adam  
batch_sizes = [16, 32]      # batch size for training cnn
```

Config_idx	CSP_Components	Depth_Multiplier	Learning_Rate	Batch_Size	Avg_Accuracy baseline
1	4	1	0.001	16	0.7011
2	4	1	0.001	32	0.6949
3	4	1	0.0005	16	0.708
4	4	1	0.0005	32	0.7034
5	4	2	0.001	16	0.7003
6	4	2	0.001	32	0.7049
7	4	2	0.0005	16	0.7042
8	4	2	0.0005	32	0.7119
9	6	1	0.001	16	0.6895
10	6	1	0.001	32	0.6933
11	6	1	0.0005	16	0.6787
12	6	1	0.0005	32	0.691
13	6	2	0.001	16	0.7034
14	6	2	0.001	32	0.7034
15	6	2	0.0005	16	0.7003

Sweep 3 : without minmax scaler, 2 cnn

csp_components = [6, 8] # n_components for CSP
depth_multipliers = [2, 4] # DepthwiseConv2D depth multiplier
learning_rates = [1e-4, 2e-3] # learning rate for Adam
batch_sizes = [8, 16] # batch size for training

Config_idx	CSP_Components	Depth_Multiplier	Learning_Rate	Batch_Size	Avg_Accuracy baseline
1	6	2	0.0001	8	0.6879
2	6	2	0.0001	16	0.6833
3	6	2	0.002	8	0.6988
9	8	2	0.0001	8	0.6802
10	8	2	0.0001	16	0.6710
11	8	2	0.002	8	0.6957

Sweep 4 : gan_augmentation_sweep.py

```
wgan_epochs_list = [100, 150]
filter_selectivity = [0.25, 1.0] # 25% highest score filter vs 100% (No filter)
aug_ratios = [0.5, 1.0] # 50% vs 100% added fake data
paradigms = ["mix", "pretrain"] # Direct Mixing vs Transfer Learning (Pre-training + fine-tuning)
```

Transfer learning didn't work well

Config_Idx	WGAN_Epochs	Filter_Selectivity	Aug_Ratio	Paradigm	Avg_Accuracy Augmented
1	100	0.25	0.5	mix	0.6079
3	100	0.25	1.0	mix	0.6310
5	100	1.0	0.5	mix	0.5916
6	100	1	.5	pretrain	0.5522
7	100	1.0	1.0	mix	0.6664
8	100	1	1	pretrain	0.5475
9	150	0.25	0.5	mix	0.6756
11	150	0.25	1.0	mix	0.6756
13	150	1.0	0.5	mix	0.6772
15	150	1.0	1.0	mix	0.6733

Sweep 5 : code_version15_10c_csp+cnn_8020_filtered_sweep.py

Gan-250 , 2 runs of CNN

- n_critic: [5, 7]
- batch_size: [16, 32]
- gen_lr: [5e-5, 1e-4] (while keeping critic_lr at 1e-4)
- latent_dim: [50, 100]

n_critic	batch_size	gen_lr	latent_dim	avg_accuracy
5	16	5.00E-05	50	0.739464
5	16	5.00E-05	100	0.731801
5	16	0.0001	50	0.681992
5	16	0.0001	100	0.697318
5	32	5.00E-05	50	0.685824
5	32	5.00E-05	100	0.727969
5	32	0.0001	50	0.727969
5	32	0.0001	100	0.689655
7	16	5.00E-05	50	0.662835
7	16	5.00E-05	100	0.689655
7	16	0.0001	50	0.704981
7	16	0.0001	100	0.704981
7	32	5.00E-05	50	0.716475
7	32	5.00E-05	100	0.731801

80- 20 split

* **CSP+CNN_150** : 80/20 split , 3x pool_multiplier, 50% augmentation,
GAN_150 , channels - p8

Code_version15_10c_csp+cnn_8020_filtered.py

Baseline CNN : Avg acc. = 0.76 +/- 0.1120

Augmented CNN : Avg acc. = 0.7548 +/- 0.1503

Subject ID	Baseline	Augmented
Subject A01	0.8214	0.8621
Subject A02	0.5517	0.5517
Subject A03	0.8966	0.8966
Subject A04	0.7241	0.7241
Subject A05	0.8611	0.9310
Subject A06	0.6552	0.5172
Subject A07	0.6897	0.6552
Subject A08	0.8912	0.9310
Subject A09	0.7586	0.7241

CSP+CNN_150 : 80/20 split , 10x pool_multiplier, 100% augmentation,
GAN_400 , avg = 0.7548 +/- 0.1503

Subject ID	Baseline	Augmented
Subject A01	0.8276	0.8621
Subject A02	0.5517	0.6207
Subject A03	0.8966	0.9655
Subject A04	0.7241	0.5862

Subject ID	Baseline	Augmented
Subject A05	0.8621	0.8966
Subject A06	0.6552	0.6207
Subject A07	0.6897	0.6897
Subject A08	0.8966	0.8621
Subject A09	0.7586	0.6897

CSP+CNN_150 : 80/20 split , 10x pool_multiplier, 100% augmentation,
GAN_250 , avg = **0.7471**

***version16 : CSP+CNN_150** : 80/20 split , 3x pool_multiplier, 50% augmentation, GAN_150, 6 cnn channels - new 10

code_version16_10newC_csp+cnn_8020_filtered.py

Baseline CNN : Avg acc. = 0.7653 +/- 0.1278

Augmented CNN : Avg acc. = 0.7854 +/- 0.1246

Subject ID	Baseline	Augmented
Subject A01	0.8621	0.8621
Subject A02	0.5862	0.7241
Subject A03	1.0000	1.0000
Subject A04	0.6207	0.6207
Subject A05	0.7821	0.8276
Subject A06	0.6107	0.5862
Subject A07	0.8276	0.8621
Subject A08	0.8276	0.8621
Subject A09	0.7453	0.7241

***version16 : CSP+CNN_150** : 80/20 split , 3x pool_multiplier, 50% augmentation, GAN_150 , 6 cnn channels - new 10

code_version16_10newC_csp+cnn_TE_filtered.py

Baseline CNN : Avg acc. =

Augmented CNN : Avg acc. =

CSP+CNN_150 : 10 fold Cross Validation 10x multiplier, 100% aug..
GAN_150 , aug avg = 0.6630
Baseline = 70 %

Subject ID	Baseline	Augmented
Subject A01	0.7405	0.7467
Subject A02	0.6676	0.6662
Subject A03	0.8681	0.8467
Subject A04	0.6524	0.5962
Subject A05	0.7638	0.7143
Subject A06	0.5210	0.4919
Subject A07	0.5686	0.5052
Subject A08	0.8543	0.8143
Subject A09	0.6610	0.5852